

# Thailand Semiconductor Supply Chain Risk Assessment

## Assessing Thailand's Exposure to Taiwan Semiconductor Supply Chain Disruption

One-Page Project Summary | Methakan Bualuang | Industrial Engineering, Chulalongkorn University | June 2026

|                                                                          |                                                                                            |                                                                         |                                                                                   |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>43.6%</b><br>Taiwan proxy dependency in 2024<br>Up from 22.8% in 2014 | <b>USD 8.35B</b><br>Base-case semiconductor-linked exposure<br>Across four Thai industries | <b>USD 5.77B</b><br>Electronics exposure<br>~69% of total base-case RAR | <b>USD 3.34B</b><br>Severe scenario impact<br>40% disruption severity stress case |
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### Problem

Thailand's electronics, automotive, HDD/storage, and home-appliance industries depend on imported integrated circuits. This project measures how exposed these industries are to Taiwan-related semiconductor supply disruption using public trade data and transparent risk modeling.

### Research Question

How exposed are Thai industries to Taiwan-related semiconductor supply chain disruption, and which industries face the highest potential impact?

### Methodology

- TPDI:** measures Thailand HS8542 import dependency on Other Asia, nes, used as a Taiwan/Chinese Taipei proxy.
- Revenue-at-Risk:** converts trade dependency into industry-level semiconductor-linked exposure.
- Scenario Analysis:** applies 10%, 20%, and 40% disruption severity assumptions to base-case exposure.

### Key Limitations

- Other Asia, nes is a proxy, not an official Taiwan-only bilateral figure.
- RAR and scenario impacts estimate exposure, not certain losses or forecasts.
- Industry chip-share assumptions are sensitivity inputs, not firm-level BOM data.

### Tools and Outputs

Python, pandas, Plotly, Excel, Jupyter Notebook, UN Comtrade/WITS trade data. Outputs include cleaned datasets, notebooks, source code, interactive visualizations, final report, and recommendations.

### Industry Exposure Summary

| Industry        | RAR Base | Severe | Risk Note                     |
|-----------------|----------|--------|-------------------------------|
| Electronics     | 5.77     | 2.31   | Dominant aggregate exposure   |
| Automotive      | 0.92     | 0.37   | High bottleneck risk          |
| HDD / Storage   | 0.87     | 0.35   | Critical controller chips     |
| Home Appliances | 0.78     | 0.31   | Moderate but rising intensity |

**Note:** RAR Base and Severe values are in USD billions. Severe = 40% disruption severity applied to base-case Revenue-at-Risk.

### Recommended Actions

- Build strategic inventory buffers based on bottleneck risk and component criticality, prioritizing chips with high production impact, long lead times, and low substitutability.
- Create a semiconductor supply-chain risk monitoring dashboard.
- Diversify suppliers where technically feasible, especially mature-node chips.
- Run annual semiconductor disruption scenario-planning exercises.
- Strengthen semiconductor-adjacent capabilities such as OSAT, advanced PCB, power electronics, and precision components.

